Today

- Simulation based inference

Simulation based inference (SBI)

- Basic idea
- Simulate model with different parameter values
- Compare simulations to data to infer the parameter value of the data
- Back in the 90s, ecologists working on IBMs invented this idea as “pattern oriented modeling”

Bayesian SBI

- Draw parameter values from prior
- Simulate
- Does it look like the data?
- If so, add to posterior
- Approximate Bayesian Computation (ABC)
- Neural Bayesian Computation

Bayesian SBI

Rejection ABC

Algorithm

for i in 1:n

Draw parameter θ from prior

Simulate data from model (e.g. IBM)

Compute summary statistic S_{sim}

Compute distance d between S_{sim} and S_{obs}

If d < tolerance:

accept θ

Bayesian SBI

- Summary statistics
  - data itself
  - mean
  - variance
  - max
  - final value

Bayesian SBI

- Rejection ABC
  - independent proposals for theta
  - wasteful
- ABC-MCMC
  - proposals for theta in high probability regions
- ABC-SMC
  - sequential Monte Carlo
  - reduce tolerance sequentially
  - “narrowing in”

Neural SBI

- Instead of choosing summary statistics (features), neural network learns which features of the data are informative
- Supervised learning
  - (theta, simulated data) pairs
- Train neural network to predict theta given data